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NANYANG JUNIOR COLLEGE
JC 2 Preliminary Examination
Higher 2

CANDIDATE NAME: **ANSWERS**

CLASS:

BIOLOGY

9744/02

Paper 2 Structured Questions

13 September 2022

Candidates answer on the Question Paper.

No Additional Materials are required.

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do no use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	9
2	11
3	8
4	10
5	10
6	10
7	11
8	11
9	10
10	10
Total	100

This document consists of **25** printed pages.

[Turn over

Answer **all** the questions.

1 The cell surface membrane has a fluid mosaic structure.

(a) Describe what is meant by the term *fluid mosaic*.

phospholipid (and protein) molecules, move about / diffuse / AW ;

protein (molecules), scattered / AW ; A different proteins present

[2]

(b) In 1934, the biologists Davson and Danielli published their suggestion for the structure of the cell surface membrane, as shown in Fig. 1.1.
They suggested that the membrane was a phospholipid bilayer with a layer of hydrophilic protein on both surfaces.

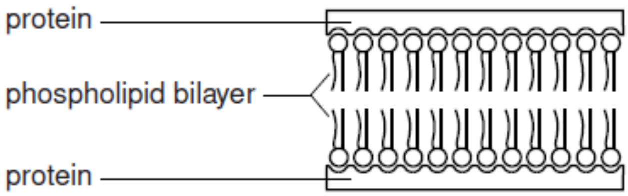


Fig. 1.1

State **one** way in which the Davson-Danielli structure is similar to the fluid mosaic structure **and one** way in which it differs from the fluid mosaic model.

similarity

(contains) phospholipid (bilayer);

A detail of orientation of phospholipid

A lipid bilayer

(contains) protein ;

difference

difference (look for ora)

(Davson Danielli) layer(s) of protein / protein only on outside ;

(fluid mosaic) ref. to proteins, in different locations discrete / different

types / named or described ;

(fluid mosaic) presence of cholesterol (molecules) ;

[2]

Fig. 1.2a is a diagram of the structure of a protein channel for ions in a cell surface membrane. Fig. 1.2b shows the channel when open and Fig. 1.2b shows the same channel when closed.

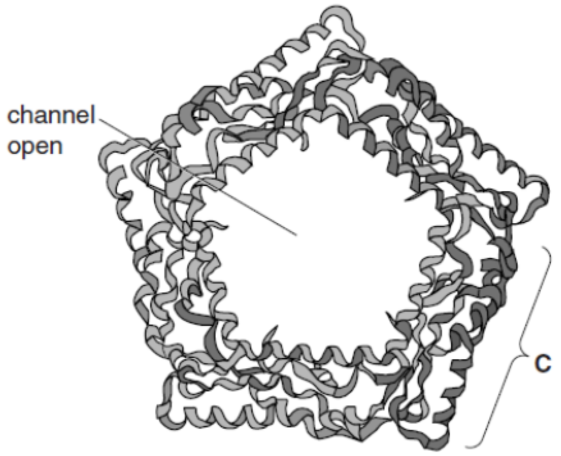


Fig. 1.2a

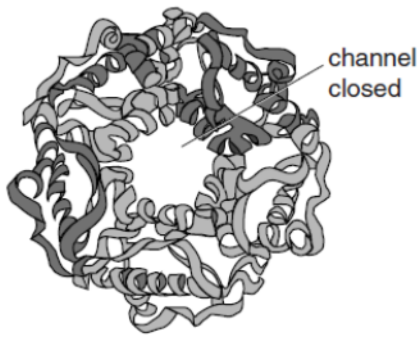


Fig. 1.2b

(c) **(i)** Explain why a channel protein is needed for ions to pass across a cell membrane.

ions are, charged / water-soluble ; A hydrophilic

unable to pass, through hydrophobic core / hydrophobic (fatty acid) tails of, phospholipid bilayer / phospholipids(s) ;

(channel of) protein lined with amino acids with, hydrophilic / polar, R groups / side chains ; A hydrophilic channels

[2]

(ii) Channel proteins are examples of transmembrane proteins. The polypeptides are held together and also interact with phospholipids in the membrane.

Suggest how the polypeptides are held together and how they interact with phospholipids.

bonds must be named in the correct context of maintaining 4° structure and interactions with phospholipids

polypeptides held together

bonds between, R groups / side chains ;

two named bond types ; from ionic / hydrogen / hydrophobic interactions / disulfide / van der Waal's forces; I peptide bond

polypeptides interact with phospholipids

(regions with) hydrophilic / charged / polar (R groups / side chains, of) amino acids interact with, phosphate / hydrophilic head , of phospholipid ;

(regions with) hydrophobic / non-polar (R groups / side chains, of) amino acids interact with, fatty acid / hydrocarbon / hydrophobic, tails / chains ;

further detail of named bond ;

[3]

[Total: 9]

2 **(a)** Table 2.1 shows some features of four biological molecules that are all polymers. Complete Table 2.1 by using a tick to indicate the features that apply to each polymer.

Table 2.1

feature	amylopectin	RNA	polypeptide	phospholipid
synthesised from amino acid monomers			✓	
polymer is branched	✓			
contains nitrogen		✓	✓	✓
can be found in both animals and plants		✓	✓	✓

[4]

Fig 2.1 is a diagram of a glycogen molecule.

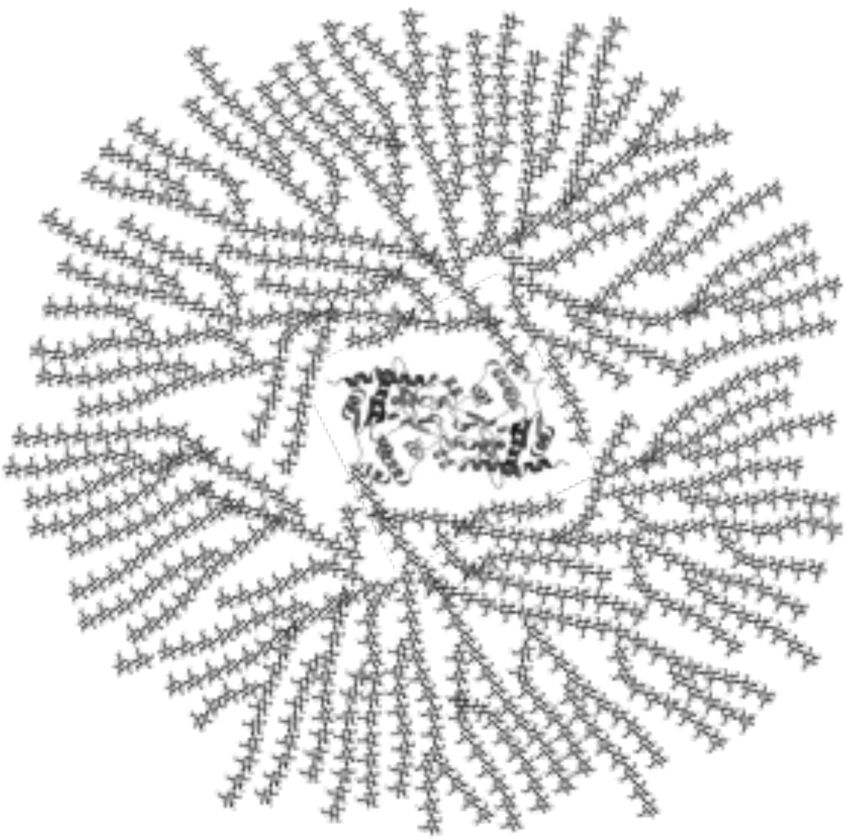


Fig. 2.1

(b) Explain how the structure of glycogen is related to its role in living organisms.

Glycogen consists of α-glucose residues linked by α-1,4-glycosidic bonds and α-1,6-glycosidic bonds where branch points occur;

Energy storage molecule + easily hydrolysed to form α- glucose, which is a respiratory substrate;

Branching results in many non-reducing ends where hydrolytic enzymes can act on;

Coils into a helical, compact structure stabilised by hydrogen bonds;

Packing of more glucose molecules in a small volume in cell;

Hydroxyl group on carbon-2 projects into the middle of the heli;

Therefore glycogen is (large and) insoluble therefore does not exert any osmotic effect;

[4]

(c) Describe how the structure of cellulose is different from the structure of glycogen.

Feature	**Glycogen**	**Cellulose**
Monomer	**α-glucose**	**β-glucose**
Bonds between monomers	**α-1,4-glycosidic bonds** and **α-1,6-glycosidic bond** at branch point	**β-1,4-glycosidic bonds**
Orientation of monomer	Adjacent glucose monomers are in the **same** orientation (No rotation)	Adjacent glucose monomers are rotated 180(with respect to each other
Overall structure	**Branched** chain polymer which coils into a **helical**, compact structure (**More extensive branching** as compared to amylopectin)	**Unbranched** chain polymer **Straight chains of β glucose run parallel to each other** Cellulose chains associate in groups to bundle into **microfibrils** and then **macrofibrils**.
Bonds that hold overall structure	**Hydrogen bonds** between hydroxyl (-OH) group of adjacent glucose residue	Intra-chain **hydrogen bonds** between adjacent molecules. Inter-chain **hydrogen bonds** between parallel chains.

[3]

[Total: 11]

3 Catalase is an enzyme that catalyses the breakdown of hydrogen peroxide, a toxic waste product of metabolism.

$$2H_2O_2 \xrightarrow{\text{catalase}} 2H_2O + O_2$$

Fig. 3.1 shows the results of an investigation into the effect of hydrogen peroxide concentration on the rate of the catalase-controlled reaction, with and without the presence of two different inhibitors.

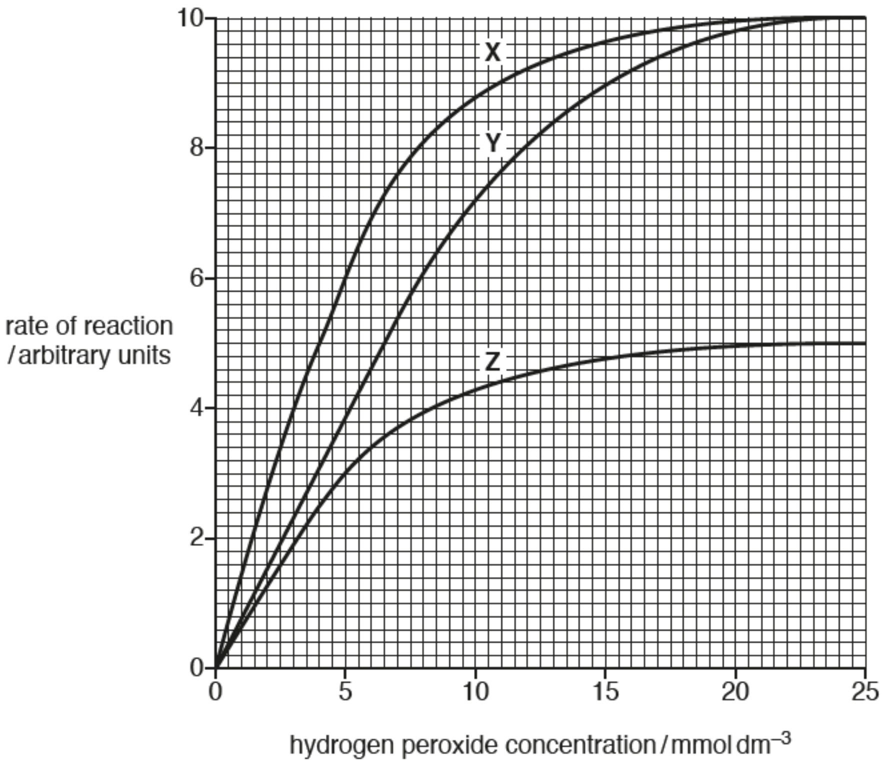


Fig. 3.1

(a) The inhibitors used in the investigation have different modes of action. Identify which of curves **X**, **Y** and **Z** are the results for:

- the reaction with the non-competitive inhibitor
- the reaction with the competitive inhibitor
- the reaction without any inhibitor.

non-competitive inhibitor ....Z..........
competitive inhibitor ....Y...........
without any inhibitor ....X...........
all three correct ;

[1]

(b) With reference to Fig. 3.1, compare the maximum rate of reaction, V_{max}, and the Michaelis-Menten constant, K_m, for curves **X**, **Y** and **Z**.

four from:

Vmax

1 X and Y same Vmax of 10 au ;

2 Vmax of, X / Y, higher than Z / ORA ; A (Vmax of), X / Y, 10 au v Z 5 au

A (Vmax of), X / Y, double the Vmax of Z

Km

3 X and Z same Km ; A Km of both is 4 mmol dm^{-3}

4 X / Z, lower Km than Y / ORA ; A Km of, X / Z, 4 mmol dm^{-3} v Y 6.5 mmol dm^{-3}

5 reference to affinity for substrate ;

[4]

(c) Account for graph Z.

(non competitive inhibition)

Inhibitor is structurally not similar to substrate/ not complementary to active site, binds at a site away from active site;

alters specific, three dimensional conformation of enzyme, distorts active site, active site no longer complementary to substrate;

fewer functional enzymes available to to catalyse reaction. E-S complexes formed per unit time decreases, hence rate of reaction is lowered.

[3]

[Total: 8]

4 Glucocorticoids (**S**), are a class of steroid hormones that bind to the glucocorticoid receptor (**GR**), and are crucial in regulation of many genes. **S** binds to **GR**, activating **GR**. Activated **GR** binds to glucocorticoid response elements (**GRE**s) within the promoter regions of target genes. This results in the recruitment of the chromatin remodelling complex, **BRG1** complex.

Fig. 4.1 shows the effect of **GR**-mediated gene expression and the effect of **BRG1** complex binding to the promoter region of the target gene.

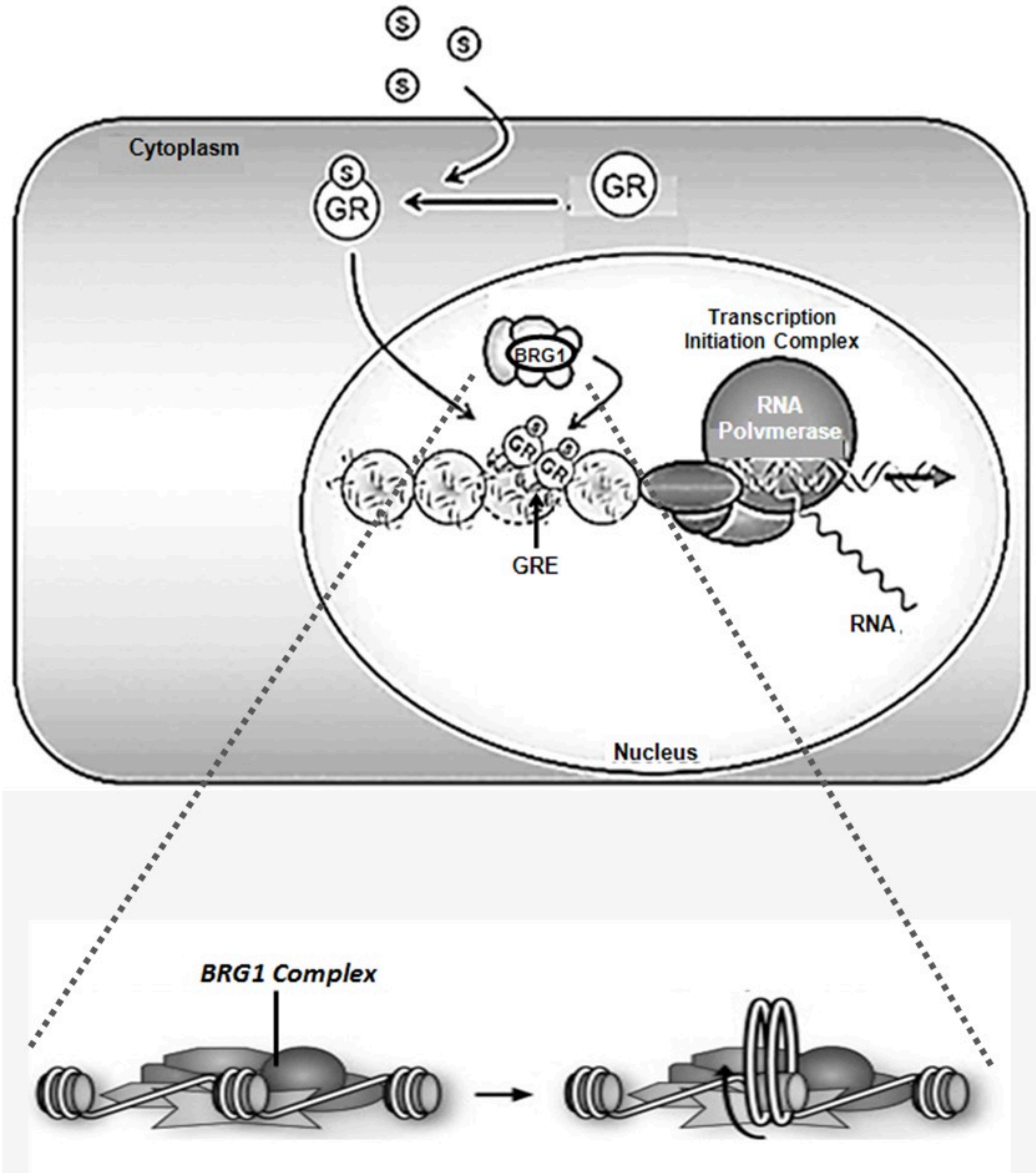


Fig. 4.1

(a) Explain how the steroid hormone is able to enter the cell.

Steroid hormone being hydrophobic/ non polar can easily diffuse through the hydrophobic core of phospholipid bilayer* of the cell surface membrane.

[1]

(b) **GR**s are known to have highly conserved regions which are structurally important for its function.
Describe two roles of **GR**s.

1. (Structure of GR) is nonpolar, allowing it to pass through nuclear envelope/ pass through nuclear pore into nucleus
2. Ref to binding sites being specific in terms of shape and charge to molecules/ substrates bound to it
3. Specific roles described (max 2)
    a. DNA binding site on GR binding to GRE at promoter region allowing transcription to proceed;
    b. Binding of steroid hormones/ GC, changing conformation of GC, allowing GC to bind to GRE;
    c. Binding of BGR1 complex, resulting in decreased compaction of DNA, allowing transcription to be upregulated/ allowing RNA polymerase to bind; there is some original answer from QB here
    d. Binding site for GR, forming a GR-GR complex, which recruits RNA polymerase and transcription factors for the formation of transcription initiation complex

[4]

(c) Levels of **GR** synthesis can be controlled translationally.
Describe how translational control can occur in eukaryotic gene expression.

Translational initiation: Phosphorylation and dephosphorylation of translation initiation factors can activate or deactivate them needed for initiate ribosome binding;

Translational repressors are proteins that bind to 5’ UTR preventing ribosome from progressing / binding to mRNA;

The longer the poly-A tail of mature mRNA, the more stable an mRNA molecule is, the longer it is translated and more polypeptides are produced.

Cytoplasmic polyadenylation: Some mRNA contain short poly A tail. Cytoplasmic polymerase catalyses cytoplasmic polyadenylation of the 3’ end of the mRNA to activate translation for temporal control;

High mRNA can be localised and concentrated in specific locations within cells to allow for spatial control of gene expression;

[3]

(d) State two differences between the cell structures used in translation in prokaryotes and eukaryotes.

In eukaryotes, 80S ribosome are used for translation (70S in chloroplast / mitochondria) whereas in prokaryotes, 70S ribosomes are used in translation;

In eukaryotes, ribosomes exist either as free ribosomes in cytoplasm or are attached to the rough endoplasmic reticulum whereas in prokaryotes, ribosomes are only found in cytoplasm;

Lack of nucleus/ chromosome lies free in cytoplasm, thus allowing simultaneous transcription and translation in prokaryotes;

[2]

[Total: 10]

5 Fig. 5.1 shows the structure of the influenza virus.

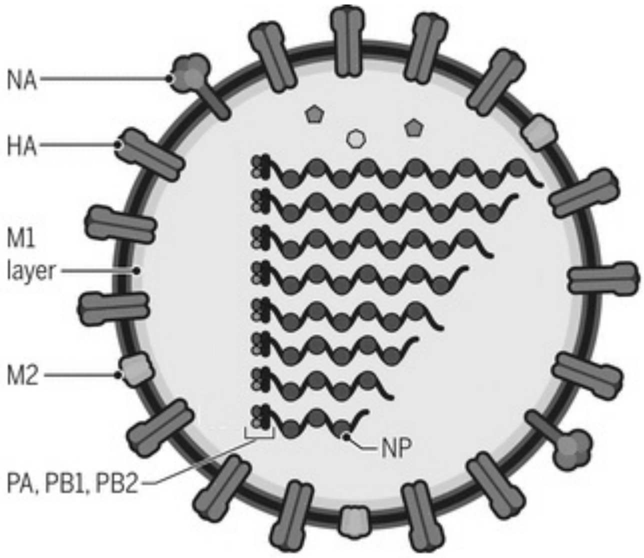


Fig. 5.1

(a) Identify the structures labelled:

label	name of structure
NA	neuraminidase
NP	nucleoproteins

[1]

(b) The RNA segments containing genes PA, PB1 and PB2 shown in Fig. 5.1 code for subunit complexes that form RNA polymerases. Explain why such RNA segments are needed by the virus.

Viral <u>RNA – dependent RNA polymerases</u> required by the virus are <u>not</u> present in the <u>host</u> cell;

They are required to synthesise positive-sense RNA, which functions as template for <u>replication of viral RNA genome</u> and are translated to <u>synthesise viral protein</u>;

[2]

(c) The diagram below shows the structure of HA and positions of amino acid where changes are frequently observed in antigenic variants of the virus.

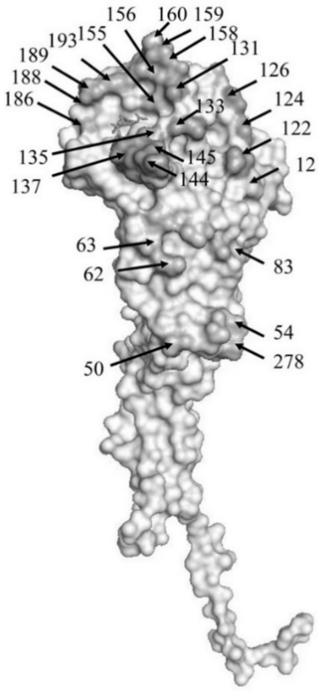


Fig. 5.2

(i) Identify the type of antigenic changes that resulted in the variants.

Antigenic drift;

[1]

(ii) Explain how the features of the virus contribute to these antigenic changes.

Single stranded RNA genome;

No backup copy to do correction or repair;

OR (RNA-dependent) RNA polymerase lacks proof-reading abiilty;

Error in viral genome replication; (or with point 8) OR RNA genome;

More reactive (due to the 2'OH group); R unstable OR Fast/high rate of replication of the virus;

Accumulate more mutations;

*Result in change in nucleotide sequence and thus amino acid sequences of glycoproteins; (Compulsory point)

(must have, e.g. 1+2+9)

[3]

(d) Contrast the reproductive cycle of influenza and HIV.

Ref to host cells and attachment stage; gp120 vs HA

Entry: Influenza enters host cells via endocytosis by invagination of cell surface membrane while HIV enters by fusing its envelope with host cell surface membrane ;

No integration of influenza viral genome vs integration of HIV genome into host cell's DNA as provirus (ds viral DNA);

Synthesis of viral components

Negative-sense RNA used as template for synthesis of positive-sense RNA strands by viral RNA-dependent RNA polymerase in nucleus

Vs

Viral RNA is reverse transcribed to from a double stranded viral DNA by reverse transcriptase. Integration of viral DNA, as a provirus, into host cell's DNA by integrase

[3]

[Total: 10]

6 Fig. 6.1 shows a stage in the mitotic cell cycle in an animal cell.

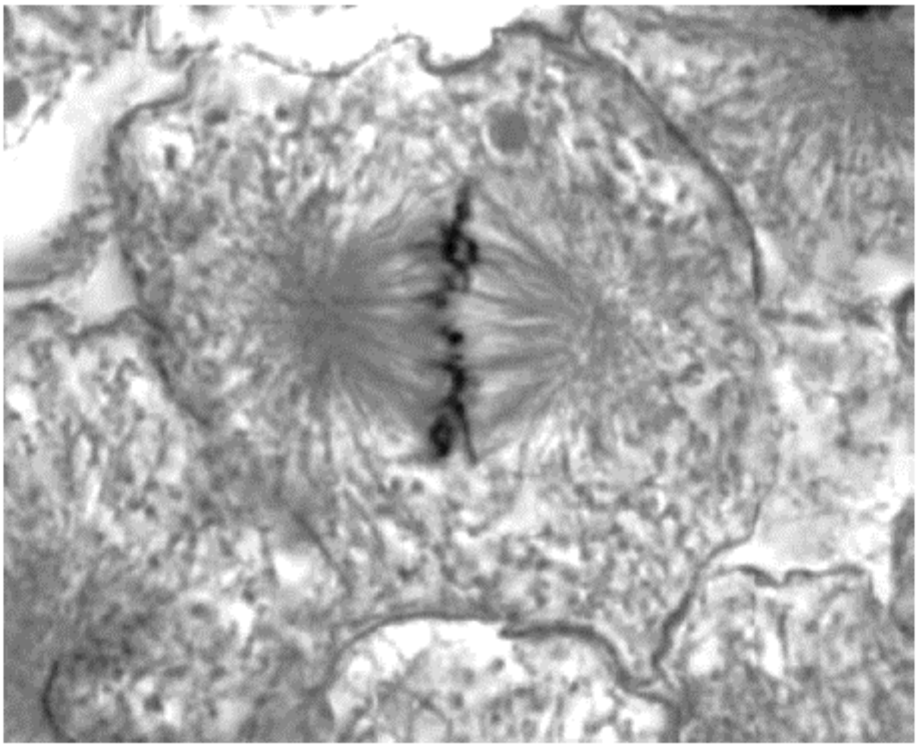


Fig. 6.1

(a) With reference to Fig. 6.1,

(i) identify the stage of mitosis;

Metaphase;

[1]

(ii) state two features which are characteristic of this stage.

Chromosomes line up at the equator of the cell/ metaphase plate.

Centromere/kinetochore attached to spindle fibres/microtubles from the centrioles.

Centrioles reach/located at poles of the cell.

[2]

(b) Distinguish between the terms *haploid* and *diploid*.

Haploid refers to only one set of chromosome being present in a cell,

Whereas diploid refers to cell having 2 sets of chromosomes,

OR

Haploid condition consists of one member of each pair of homologous chromosome present.

Diploid condition consists of 2 sets of chromosomes, one set derived from each parent.

[2]

(c) Explain the importance of mitosis in organisms.

maintains / same, genetic stability / number of chromosomes/ two sets of chromosomes / diploid / 2n /

produces daughter cells that are genetically identical to parent cells;

replacement of cells / tissue repair

growth / increase in cell numbers ;

asexual reproduction;

[3]

(d) In many multicellular organisms, such as mammals, the time taken for the mitotic cell cycle varies considerably between different tissues but is very carefully controlled in each cell.

Suggest the importance of this control in mammals.

Prevent tumour/ cancer formation due to uncontrolled cell division.

Only cells that are needed / functions are needed will be produced

Allows for coordination of growth / limiting growth ;

[2]

[Total: 10]

7 A group of scientists discovered a new species of forest cockroach near their campus. Female cockroaches of this species are homogametic and the species follows the XO sex determination system.

Upon analysis of the samples they had collected, it was observed that there are three different eye colours in the species. These eye colours are determined by two genes, Gene B and Gene R, which are found on different chromosomes. One of the genes is present on the X chromosome while the other is found on an autosome. Fig 7.1 shows the formation of eye pigment in this forest cockroach species.

$$\text{Blue pigment} \xrightarrow{\text{Enzyme B}} \text{Purple pigment} \xrightarrow{\text{Enzyme R}} \text{Red pigment}$$

Fig. 7.1

The dominant B allele converts a blue pigment into a purple pigment while the dominant R allele converts the purple pigment into a red pigment.

(a) (i) State the pattern of inheritance of the eye colour in the new species of forest cockroach.

Epistasis;

[1]

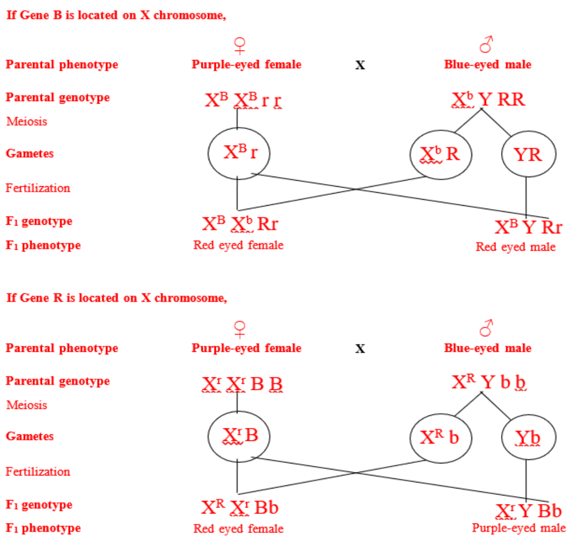
(ii) A true-breeding blue-eyed male was crossed to a true-breeding purple-eye female. All the F1 offspring from this cross had red eyes.

Deduce the gene that is located on the X chromosome and explain how you derived your answer.

Gene B;

Explain: If gene R is on X chromosome, F1 phenotypes would show red eyed females AND purple eyed males;

[2]



(iii) Draw a genetic diagram showing the expected ratio of phenotypes when the F1 progeny are crossed.

Parental phenotype	Red eyed, female	x	Red eyed, male
Parental genotype	X^BX^bRr	x	X^BYRr
After meiosis, Gametes	X^BR X^Br X^bR X^br	RrBb	X^BR X^Br $0Y$ $0R$

	X^Br	X^BR	X^br	X^bR
0r	X^B0rr Purple, male	X^B0Rr Red, male	X^b0rr Blue, male	X^b0Rr Blue, male
0R	X^B0Rr Red male	X^B0RR Red, male	X^b0Rr Blue, male	X^b0RR Blue, male
X^BR	X^BX^BRr Red, female	X^BX^BRR Red eye, female	X^BX^bRr Red eye, female	X^BX^bRR Red eye, female
X^Br	X^BX^Brr Purple, female	X^BX^BRr Red, female	X^BX^bRr Purple, female	X^BX^brr Purple, female

F2 phenotypic ratio 3 red eyed male : 1 purple eyed male : 4 blue eyed male: 6 red eyed female: 2 purple eyed female

[4]

(b) (i) Another species of the forest cockroach also has three eye colours but the expression of eye color in this species of forest cockroach is only controlled by one gene C found on an autosome.

It was found that a cross between true-breeding blue-eyed and red-eyed individuals produced all purple-eyed F1 progeny.

If the purple-eyed F1 progeny are mated with each other, deduce the expected F2 phenotypic ratio.

1 blue : 2 purple : 1 red (Incomplete dominance)

[1]

(ii) 188 offspring were obtained in the F2 generation and the phenotypes of the offspring were as follows:

40 cockroaches with blue eyes

100 cockroaches with purple eyes

48 cockroaches with red eyes

A χ^2 (chi-squared) test was performed to test the significance of the difference between the observed and expected results. The calculated χ^2 = 1.45.

Part of the critical values of the chi-squared distribution is shown below.

Degrees of freedom	Probability		
	0.10	**0.05**	**0.01**
1	2.71	3.84	6.64
2	4.69	5.99	9.21
3	6.25	7.82	11.35
4	7.78	9.49	13.28

Explain whether your calculated χ^2 value in support the expected phenotypic ratio that you have obtained in **(b)(i)**.

Probability = 0.10 < p;

At 0.05 level of significance, since calculated (1.45) is less than critical value of 5.99;

Difference between observed and expected result is not significant and is due to chance;

[3]

[Total: 11]

8 **(a)** Photosynthesis is a complex process involving a light dependent stage and a light independent stage.

(i) Name the products of the light dependent stage that are needed in the light independent stage.

ATP;

Reduced NADP;

[2]

(ii) Describe the role of chlorophyll b in photosynthesis.

Accessory pigment found in the light harvesting complex ;

Absorbs / harvests light energy ; Ignore trap / capture

Pass energy to neighbouring accessory pigments until it accumulates and reaches special chlorophyll a in reaction centre

AVP ; e.g. idea of extending the range of light wavelengths absorbed / absorbs wavelengths not absorbed by primary pigment

[2]

A student carried out an experiment to investigate the effect of light intensity and light wavelength on the rate of photosynthesis.

- An aquatic plant, *Elodea canadensis*, was put into a beaker containing sodium hydrogencarbonate solution as a source of carbon dioxide.
- To minimise changes in temperature, an LED lamp was used as a source of light.
- The lamp was switched on and the number of bubbles released by the aquatic plant in 1 minute was counted.
- The lamp was placed at seven different distances from the beaker to change light intensity.
- Five replicates were carried out at each lamp distance.
- All other variables were controlled.

(b) The student calculated the light intensity for each distance (d) using $\frac{1}{d^2}$.
Table 8.1 shows the calculated light intensities for each distance.

Table 8.1

distance between plant and lamp / m	light intensity / $\frac{1}{d^2}$
0.025	1600
0.050	400
0.100	100
0.150	44
0.200	25
0.250	16
0.300	11

Complete Table 8.1 by calculating the light intensity for distance 0.100m. [1]

(c) At each distance from the lamp, the experiment was repeated using a red filter in front of the lamp to give a different wavelength of light. The experiment was repeated using a blue filter and then using a green filter. Each filter transmitted the same light intensity.

The student calculated the mean rate of bubble production as a measure of the rate of photosynthesis.

Fig. 8.1 shows a graph of the results.

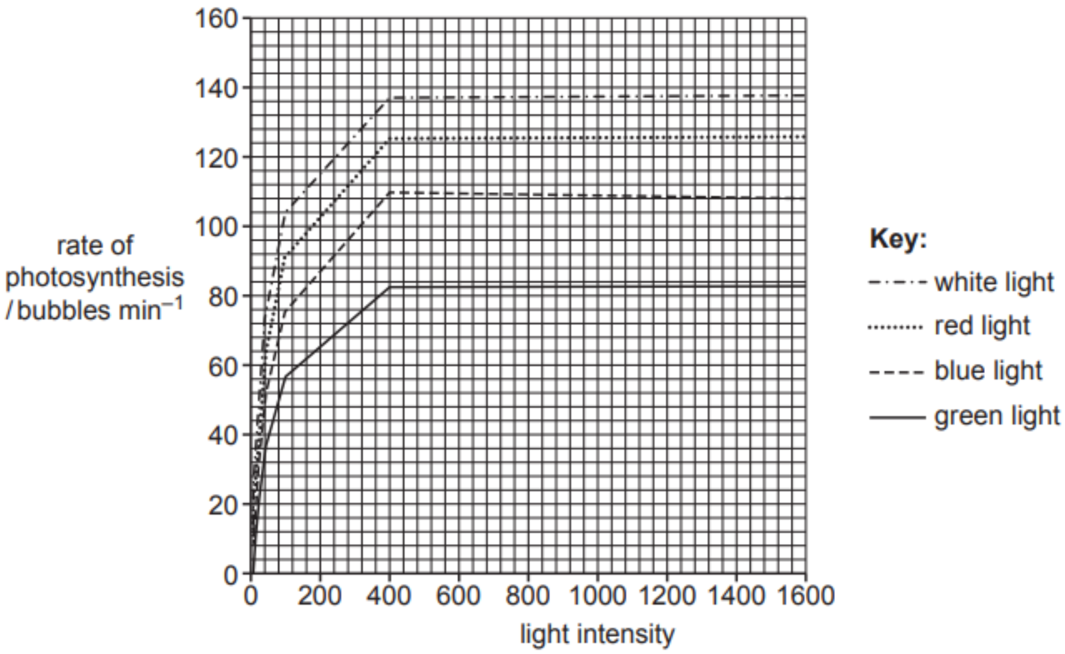


Fig. 8.1

(i) With reference to Fig. 8.1:

- state the range over which light intensity is the limiting factor
- explain why light intensity above this range is not limiting the rate of photosynthesis.

0 – 400 ; @ up to 400 ignore units

Any **two** from:

temperature / concentration of carbon dioxide, becomes the limiting factor ;

(at a lower than optimum temperature)

less kinetic energy so fewer collisions between rubisco and CO2 OR

a lower temp will limit the rate of (named) enzyme-controlled reactions ;

(with a lower than optimum CO2 concentration) less fixation of CO2 / AW ;

all, enzymes / processes, in photosynthesis, already at, highest rate / optimum OR maximum amount of light is being absorbed (by the pigments) ;

[3]

(ii) At a light intensity of 1600, explain why different colour filters result in different rates of photosynthesis.

Any three from:

different wavelengths of light absorbed by different pigments ;

red light is absorbed the most ;

green light is, absorbed less / reflected ;

shorter the wavelength the greater the energy / red light has more energy (than blue light) ; ora

the more, light / energy, absorbed the more, light dependent reaction / photophosphorylation, occurs ; ora

[3]

[Total: 11]

9 (a) The European eel, *Anguilla anguilla*, is a fish. The sizes of eel populations tend to remain relatively stable despite eels producing large numbers of offspring.

Suggest **two** reasons why the population sizes of eels tend to remain relatively stable.

Any two from:

Predation / fishing;

Food qualified e.g. competition or limited amount ;

Disease;

AVP e.g. idea that birth and death rate roughly equal;

[2]

(b) Explain what is meant by the theory of evolution.

Changes in allelic frequency over (long periods of) time;

Ref to natural selection / selective advantage for survival / survival of the fittest;

Ref to variation (hence nature has a choice);

@ descent with modification

[2]

(c) The generation time of a species is the mean (average) time from one generation (parents) to the next generation (offspring). For example, the generation time of humans is about 25 years.

Fig. 9.1 shows a graph of the relationship between the rate of evolution and the generation time for a wide range of different species.

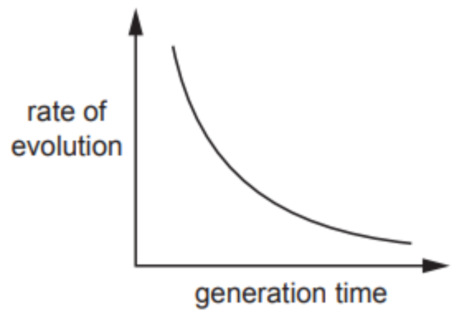


Fig. 9.1

Describe **and** explain the relationship shown in Fig. 9.1.

as the generation time increases, the rate of evolution decreases / relationship is inversely proportional / AW ; ora

not linear / exponential / decreasing gradient ;

(longer generation time) so less, reproduction / DNA replication ; ora ref. to fewer mutations ; ora

idea of (so) less chance of evolution ; ora *mp5 must be linked to mp3 or mp4*

[3]

(d) The tuatara, *Sphenodon punctatus*, is a reptile that is native to New Zealand. It is found nowhere else in the wild.

Fig. 9.2 shows a tuatara.

Fig. 9.2

Tuataras have a slow growth rate and can live for over one hundred years. Fossil evidence shows that there has been little morphological change in the tuatara over the last 200 million years. This is a much lower rate of evolution than would be expected from the generation time of this species.

Suggest why the tuatara has remained largely unchanged over the last 200 million years.

well adapted to their environment ;

environment / selection pressures, stayed, constant / stable ; ora Rej: no selection pressures

mutations qualified ; e.g. fewer / not selected for / low rate

AVP ; e.g. no natural predators / cannot migrate / no separation of populations (for speciation to occur)

[3]

[Total: 10]

10 Dengue viruses (DENV) are responsible for millions of infections each year in tropical and subtropical areas of the world. According to the World Health Organization, dengue incidence has increased significantly over the past 50 years, turning this infection into the most important mosquito-borne disease in the world and a global health challenge. Fig. 10.1 shows the general global distribution of dengue fever in the year 2005.

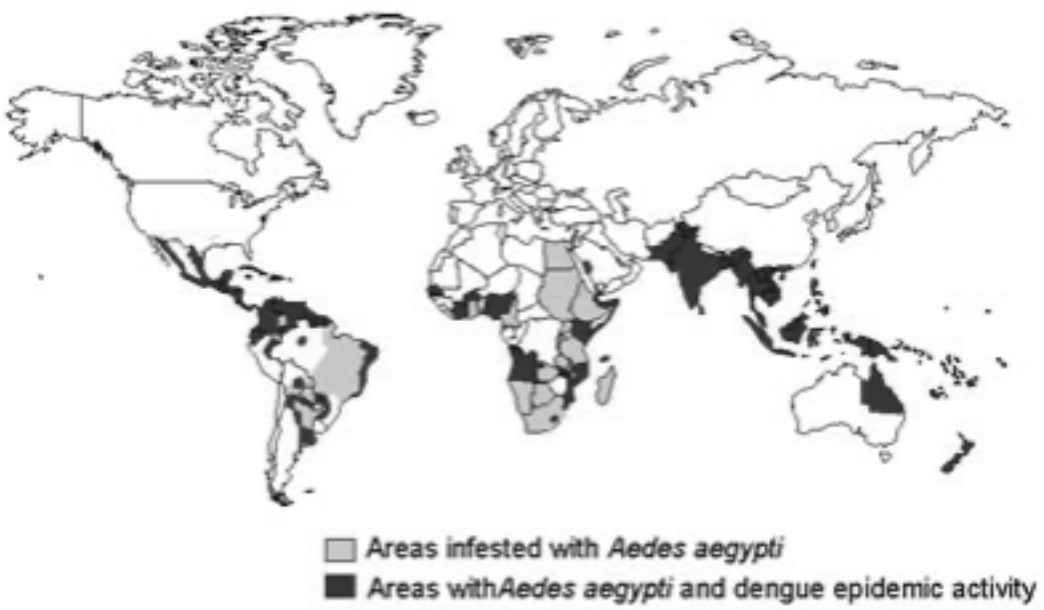


Fig. 10.1

(a) (i) Climate is one of the important factors that affects the distribution of dengue.

Predict and justify the expected distribution of dengue by the end of the 21st century.

1. Geographical range of Aedes mosquito (vector) and dengue will expand towards the two poles/ to areas infested with mosquitoes;

2. As global temperature is predicted to rise (by 4°C)/ global warming/ climate change due to (increase in greenhouse gas emission);

3. resulting in favourable temperatures for mosquito breeding/ suitable breeding places in the temperate regions/ increased viral load/ higher rate of DENV replication;

[2]

(ii) Symptoms of dengue such as fever usually develop within 4 to 7 days after being bitten by an infected mosquito and is often associated with joint pain.

Explain how DENV may cause these symptoms.

1. Virus stimulates/ activates innate/ non-specific immune response/ immune cells (e.g. dendritic cells) in the first four days → fever? ;

2. Cytokines and chemokines released by macrophages increased permeability of blood vessels resulting in inflammation, causing pain;

3. Pyrogen released by activated macrophages leads to rise in systemic body temperature resulting in dengue fever;

**This question focus is actually more infectious diseases, specifically on innate immune system*

[2]

(b) Outline the roles of T lymphocytes in the adaptive immune response after an infection with DENV.

CD4 T cells activated by APCs to form T helper cells ;

T helper cells secrete cytokines to activate B cells and CD8 T cells ;

CD8 T cells form cytotoxic T cells which kill infected cells ;

Via perforins and granzymes ;

[4]

(c) To some extent the range of the *Aedes* mosquito has also followed human expansion. Explain how this may be true.

ref. human activity provide viable habitats e.g. stagnant water for Aedes to thrive

In colder latitudes or altitudes, urbanisation / large cities provide a warmer habitat

[2]

[Total: 10]